

3 During aerobic respiration, cells respire substrates such as glucose to produce ATP.

Some events that occur during aerobic respiration are:

- The respiratory substrate breaks down into smaller and smaller molecules. These series of reactions are described as catabolism.
 - Coenzymes take part in various reactions. In some reactions, coenzymes are reduced or oxidised.
 - Carbon dioxide is released.
- (a)** Aerobic respiration occurs in four successive stages: glycolysis (**G**), link reaction (**LR**), Krebs cycle (**KC**) and oxidative phosphorylation (**OP**).

Complete Table 3.1 to show which events occur in each stage of aerobic respiration. Use a tick (✓) to show that the event does occur or a cross (✗) to show that the event does **not** occur.

Table 3.1

event	stage			
	G	LR	KC	OP
catabolism				
coenzyme is reduced or oxidised				
a coenzyme forms a covalent bond with a respiratory intermediate				
carbon dioxide is released				

[4]



- (b) A new hand-held technological device shows the main type of respiratory substrate being used in the cells of a person.

The device consists of a carbon dioxide sensor and air-flow meter. The person inhales through the device for a fixed time and then exhales into it.

The device calculates the respiratory quotient (RQ) value to show whether the cells are mainly respiring carbohydrates or lipids.

- (i) Explain how the device calculates the RQ value **and** how this shows whether the cells are mainly respiring carbohydrates or lipids.

.....

.....

.....

.....

.....

.....

.....

..... [4]

- (ii) State the difference in the relative energy values of carbohydrates and lipids as respiratory substrates, **and** explain the reasons for the difference.

.....

.....

.....

.....

.....

..... [3]

[Total: 11]